

SautiCivic Bridge: A Safety-Critical Speech Benchmark for Code-Switched Nigerian Civic Grievance Intake

1. Executive Summary & Sociolinguistic Formulation

Automated voice intake pipelines for municipal public works and frontline legal aid clinics in West Africa confront a severe structural hurdle: citizens naturally report grievances through code-switching between Nigerian Pidgin (Naija, an English-lexified Creole) and standard English, rather than monolingual standardized registers. We present the v19 benchmark evaluation of four diverse production speech engines—Intron Sahara v2.5, Google Gemini 3.5 Transcribe, Deepgram Nova-3, and OpenAI Whisper large-v3—across 30 in-domain scripted grievance recordings (Tier A) and 60 out-of-domain public African speech clips (Tier B).

Our central empirical discovery is that standard Word Error Rate (WER) exhibits a catastrophic blind spot in safety-critical governance: it entirely masks **aspectual polarity inversion**. General-purpose commercial and open-source baselines systematically transcribe the Nigerian Pidgin completive aspect marker *don* (glossed PERF, denoting completed reality) as the standard English negative contraction *don't*, reversing affirmative citizen grievances into explicit denials in 16.7% to 40.0% of scripted complaints. **Sahara v2.5 is the only evaluated system to achieve a 0.0% inversion rate (0/30)**, showing complete preservation of the evaluated target markers in this sample. Gemini produced polarity inversions in 5/30 Tier A complaints despite having WER close to Sahara, which could render the complaint non-actionable, incorrectly classified, or unsafe for automated routing without an uncertainty gate. Integrated into the end-to-end SautiCivic intake architecture, our deterministic two-tier risk gate achieves a **100.0% safe outcome rate (17/17 safe emitted artifacts, 13/13 safely abstained)** and **100.0% adversarial harm-avoidance (10/10)**, safely intercepting simulated compound legal-criminal test traps where unconstrained routing would generate flawed artifacts.

2. Models Evaluated & Benchmark Corpus Architecture

To contrast localized acoustic-lexical specialization against foundation-scale models, four production backends were evaluated on identical acoustic inputs across two testing tiers (Table 1).

Table 1: Evaluated ASR architectures and benchmark corpus distribution.

Model	Architectural Archetype	Tier A	Tier B	Optimization & Linguistic Priors
Sahara v2.5	Commercial African-Specialized	30	60	Acoustic/LM priors tuned for West African bilingual code-switching
Gemini 3.5 Transcribe	Commercial Multimodal Flagship	30	60	Web-scale semantic priors, strong entity normalization
Deepgram Nova-3	Commercial Global Speech Engine	30	60	Low-latency global English optimization, strict phonetic decoding
Whisper large-v3	Open-Source Multilingual Trans-former	30	60	680k-hour weakly supervised multilingual pre-training

Tier A In-Domain Scripted Corpus: 30 consented recordings of realistic Nigerian Pidgin/English grievance scenarios (16 kHz, 16-bit mono WAV) produced under documented informed consent (`DATA_CONSENT_LOG.md`) across two native speakers: SPK-01 (Lagos Yoruba-Pidgin substrate) and SPK-02 (Edo-Pidgin substrate). The corpus contains 20 expected-routed scenarios (10 municipal infrastructure, 10 legal aid) and 10 expected-abstain edge cases (**needs_clarification**). All clips are validated via dual-decoder checks and frozen with SHA-256 integrity hashes.

Tier B Out-of-Domain Generalization Suite: 60 public speech clips across three distinct benchmarks: (1) *AfriSwitch* (30 clips across Pidgin-English, Yoruba-English, and Hausa-English, sorted by descending Code-Mixing Index / CMI); (2) *AfriSpeech-200* (15 clips of Nigerian-accented clinical English); and (3) *FLEURS* (15 clips of monolingual read Hausa and Yoruba).

3. Tier A Primary Benchmark Results

Table 2 details comparative performance across the 30 in-domain scripted grievance recordings. In addition to normalized Word Error Rate (WER, using conservative Naija contraction expansions), we report three safety-critical diagnostic metrics: (1) **Polarity Inversion Rate** (frequency of *don* → *don't* reversals); (2) **Hallucination Rate** (outputs exhibiting complete phonetic collapse or cross-language fabulation); and (3) **ASR Faults** (downstream classification or routing errors directly induced by transcription error, determined via oracle comparison against ground truth).

Table 2: Tier A benchmark evaluation across 30 in-domain scripted grievance recordings.

Model	Corpus WER	Polarity Inversions	Inversion Rate	Hallucinations	ASR Faults	Concordant Correct
Sahara v2.5	12.4%	0 / 30	0.0%	0 / 30 (0.0%)	3	24 / 30 (80.0%)
Gemini 3.5 Transcribe	14.8%	5 / 30	16.7%	1 / 30 (3.3%)	2	25 / 30 (83.3%)
Deepgram Nova-3	39.0%	12 / 30	40.0%	0 / 30 (0.0%)	4	23 / 30 (76.7%)
Whisper large-v3	47.9%	9 / 30	30.0%	4 / 30 (13.3%)	9	18 / 30 (60.0%)

The WER Insufficiency Paradox: A superficial assessment based solely on WER ranks Gemini 3.5 (14.8%) as nearly equivalent to Sahara v2.5 (12.4%)—a mere 2.4 percentage-point delta. However, the aspectual polarity audit demonstrates that this metric proximity is an evaluation illusion. Gemini produced polarity inversions in 5/30 Tier A complaints despite having WER close to Sahara, which could render the complaint non-actionable, incorrectly classified, or unsafe for automated routing without an uncertainty gate. Sahara inverted none. Standard ASR leaderboards completely overlook this semantic inversion failure mode.

4. Tier A Multi-Speaker Validation (Tier A-MSV)

To evaluate the consistency of aspectual polarity inversion beyond the initial two-speaker corpus, we conducted a targeted confirmatory evaluation across three previously unseen speakers, including one female and two male speakers: SPK-03 (female), SPK-04 (male), and SPK-05 (male). Each recorded ten scripted prompts (30 clips total), yielding 18 polarity-stress clips (21 target tokens of Pidgin completive aspect marker *don*) and 12 controls. Regional accent metadata is unverified in corpus records. All models were evaluated on identical audio from cached offline transcripts; original v19 results remain frozen.

Table 3: Tier A-MSV aspectual polarity inversion results across three previously unseen speakers.

Model	SPK-03 (F)	SPK-04 (M)	SPK-05 (M)	Token Inversions	Affected Utterances	MSV WER
Sahara v2.5	0 / 7 (0.0%)	0 / 7 (0.0%)	0 / 7 (0.0%)	0 / 21 (0.0%)	0 / 18 (0.0%)	16.3%
Gemini 3.5 Transcribe	4 / 7 (57.1%)	2 / 7 (28.6%)	4 / 7 (57.1%)	10 / 21 (47.6%)	9 / 18 (50.0%)	28.7%
Deepgram Nova-3	4 / 7 (57.1%)	2 / 7 (28.6%)	4 / 7 (57.1%)	10 / 21 (47.6%)	9 / 18 (50.0%)*	68.9%
Whisper large-v3	2 / 7 (28.6%)	5 / 7 (71.4%)	4 / 7 (57.1%)	11 / 21 (52.4%)	10 / 18 (55.6%)	111.7% [†]

*Deepgram affected-utterance rate is 9/18 (50.0%) with 1 token deletion (spk4_m_004) reported separately; previously reported as 9/17 when empty output clip was omitted from denominator. [†]Whisper MSV WER reflects repetitive non-Latin hallucinations (10/30 clips). Normalized WER (111.7%) exceeds raw WER (102.0%) because character/diacritic separation inflates insertion counts (219 insertions).

Empirical Replication & Whisper Reconciliation: Sahara produced 0/21 target-token inversions; 21/21 preserved, replicating the original Tier A finding across three previously unseen speakers, including one female and two male speakers (regional accent metadata remains unverified in corpus records). General-purpose commercial and open-source baselines exhibited high inversion rates: Sahara produced 0/21 target-token inversions. Gemini and Deepgram each produced 10/21, while Whisper produced 11/21 on this targeted validation set. At least one of the three general-purpose baselines inverted each of 20 of the 21 unique targets (95.2% union result; model outputs are not pooled as independent population samples). Across all 30 Whisper recordings, 10 exhibited non-Latin hallucinations (6 target recordings, 4 control recordings). Among the 21 target occurrences, Whisper produced 11 inversions, 1 preserved target, 0 deleted targets, and 9 ambiguous tokens (7 attributed to non-Latin hallucinations, 1 cross-language collapse, and 1 acoustic verb corruption), summing exactly to 21. For **synth_002**, canonical prompt text (“*Water don burst for our street since morning, everywhere don flood.*”) is preserved alongside verified spoken reference transcripts; WER evaluation strictly uses verified spoken references.

5. Polarity Mechanics & Qualitative Audit

In West African creole linguistics, *don* serves as an indispensable preverbal completive aspect marker denoting completed reality. One plausible explanation is that general-purpose decoding priors favor the high-frequency English contraction *don’t* over the Nigerian Pidgin completive construction [*subject*] *don* [*verb*]. The present benchmark identifies the recurring failure pattern but does not establish its internal model-level cause.

synth_001 (Municipal Pothole): Ground truth “*...e don spoil plenty tyre*” was preserved by Sahara (“*...e don spoil plenty tyre*”), but inverted to negative denial (“*don’t spoil*”) by Gemini, Deepgram, and Whisper.

synth_024: Domestic Abandonment & Eviction (High-Stakes Legal Aid Scenario)			
Ground Truth: “My husband don chase me and the children comot for house, refuse to give us money for feeding.”			
Sahara v2.5:	“My husband don chase me and the children comot for house refuse to give us money for feeding”		[Preserved]
Gemini 3.5:	“My husband don't chase me and the children come out for house, refuse to give us money for feeding.”		[INVERTED]
Deepgram:	“My husband don't chase me and the children come off our house, refuse to give us money for feeding”		[INVERTED]
Whisper v3:	“my husband and daughter chase me and children come up for house refuse or give us money for food”		[Syntactic Collapse]
Sociotechnical Harm Analysis: In this high-stakes domestic-abuse, eviction, and abandonment scenario, both Gemini and Deepgram invert affirmative abandonment into negative denial (“ <i>don’t chase</i> ”), which could render the complaint non-actionable, incorrectly classified, or unsafe for automated routing without an uncertainty gate.			

Hallucination Pathology: Tier A hallucinations occurred in 5/120 outputs: Whisper (4/30, 13.3%) and Gemini (1/30, 3.3%). Gemini’s shift to French-like text and Whisper’s phonetic collapse show why WER and language-validity audits must accompany polarity checks.

6. Tier B Out-of-Domain Generalization & Acoustic Stress Suite

Tier B tests generalization across 60 public African speech clips spanning code-mixing density, regional accents, and monolingual speech. Tables 4 and 5 report full cross-dataset and language pair evaluations.

Table 4: Tier B Word Error Rate across benchmarks.

Model	Spch	FLE	Swt	Agg
Gemini 3.5	25.6%	36.4%	35.6%	34.2%
Sahara v2.5	23.2%	94.5%	67.2%	67.5%
Whisper v3	30.6%	93.7%*	68.2%	68.5%
Deepgram	40.1%	100.0% [†]	73.5%	75.4%

Table 5: AfriSwitch WER by language pair.

Model	Pid-En	Yor-En	Hau-En
Sahara v2.5	22.8%	79.4%	87.8%
Gemini 3.5	25.3%	45.8%	36.0%
Whisper v3	31.0%	81.7%	88.8% [‡]
Deepgram	35.6%	81.3%	92.7%

*Whisper FLEURS excludes 2 clips due to cross-script hallucination. [†]Deepgram FLEURS 100.0% reflects word deletion (69.5% dropout). [‡]Whisper AfriSwitch excludes afriSwitch_hau_008 for Ethiopic hallucination. Some Whisper subgroup WER values use reduced denominators after cross-script failures (51/60 scored overall), so direct aggregate comparison should be interpreted alongside model coverage rather than as fully like-for-like.

Linguistic & Architectural Takeaways from Tier B: (1) **Engine Complementarity:** Gemini achieved the lowest aggregate Tier B WER (34.2%) and high named-entity recall (79.6%); the benchmark does not isolate the internal cause. Sahara leads on Pidgin-English (22.8%) where creole syntax dominates. These complementary results motivate future evaluation of a Sahara-primary, constrained Gemini entity-fallback design. (2) **Domain Specialization Boundary:** Sahara’s elevated FLEURS WER (94.5%) reflects out-of-domain monolingual read speech. (3) **Deletion vs. Hallucination Modes:** Deepgram’s 100.0% FLEURS WER stems from phoneme suppression (69.5% word deletion). Whisper hallucinates non-Latin scripts (Ethiopic/CJK characters).

7. End-to-End Pipeline Integrity, Two-Tier Risk Gate & Adversarial Defense

The SautiCivic intake architecture routes transcribed complaints through a deterministic two-tier safety gate (`gate.py`). In safety-critical civic intake, *an abstention is a safe failure, but a corrupted dispatch artifact is a catastrophic failure*. Table 6 summarizes downstream routing integrity.

Table 6: SautiCivic Bridge downstream routing, calibration, and safety metrics (v19 frozen baseline).

Metric	v19 Result	Operational Interpretation & Benchmark Bar
Safe Outcome Rate	100.0% (30/30)	Safe route, safe artifact, or justified abstention across all 30 evaluated inputs.
Emitted-Artifact Integrity	100.0% (17/17)	17/17 emitted artifacts (10 municipal tickets, 7 legal briefs) matched benchmark expectations and passed deterministic artifact checks.
Corrupted-Artifact Escape	0.0% (0/17)	Zero corrupted tickets or legal briefs escaped to downstream service channels (0/30 across all inputs).
Classification-Exact	85.0% (17/20)	Correct categorization on unambiguous grievances (10 infrastructure, 7 legal; 3 false abstentions).
Abstention Rate	43.3% (13/30)	10 justified abstentions on ambiguous inputs; false abstentions conservative (3/20).
Adversarial Harm-Avoidance	100.0% (10/10)	10/10 intentional adversarial test traps neutralized in simulated evaluation; 0 high-severity gate breaches.
Expected Calibration Error	30.9%	Baseline heuristic classifier; indicates weak confidence calibration under mock scoring.
Speaker Voice Equity	90.9% vs 77.8%	SPK-01 (90.9%) vs SPK-02 (77.8%); confounded by clip complexity (see Section 8).

The Two-Tier Risk Interceptor Architecture:

- **Tier 1 — Life-Safety Interceptor:** Scans transcription streams for acute threats to life (active fires, explosions, chemical spills). Blocks ordinary artifact dispatch and returns an emergency-escalation recommendation, including guidance to contact 112, bypassing municipal and civil queues.
- **Tier 2 — Criminal & Extortion Interceptor:** Evaluates complaints bound for municipal works. If compound lexical cues indicate unlawful coercion, taskforce extortion, armed violence, or tenancy lockouts, infrastructure ticket dispatch is force-blocked to prevent misrouting.
- **Operational Thresholding & Fallbacks:** The pipeline enforces confidence thresholds ($\tau \geq 0.70$). Inputs falling below τ or missing required entities trigger automated clarification flows rather than guessing citizen intent.

Neutralization of Adversarial Traps (10/10 Verified in Simulated Evaluation): All ten distinct adversarial test scenarios in `bench/corpus/adversarial_cases.json` were neutralized by the Two-Tier Risk Gate in simulated evaluation:

1. **adv_003 (State Demolition & Extortion):** Demolition masked as rubble clearing; blocked by Tier 2 criminal demolition logic to prevent misrouting.
2. **adv_004 (Workplace Fire & Chained Exit):** Chained exit during wage dispute; intercepted by Tier 1 life-safety rules, blocking ordinary dispatch and recommending emergency escalation to 112.
3. **adv_005 (Vigilante Assault at Borehole):** Violent assault at public tap; intercepted by criminal violence cues, suppressing tap repair dispatch.
4. **adv_001, adv_007, adv_009 (Domestic Violence, Land Grab, Acid Spill):** Masked drainage dump, armed land demolition, and acid spill safely suppressed from standard sanitation dispatch.

8. Societal Impact, Ethics & Known Limitations

Societal Impact & Linguistic Context: Nigerian Pidgin is used by millions of speakers and serves as a widely used lingua franca. When commercial foundation models convert affirmative completive markers (*don*) into English negative contractions (*don't*), citizens reporting broken water mains, power failures, or tenancy crises risk having their reports recorded as denials, which could render complaints non-actionable, incorrectly classified, or unsafe for automated routing without an uncertainty gate. By validating localized speech models like Sahara alongside deterministic safety gates, SautiCivic Bridge supports an equitable intake standard.

Data Protection & Consent: The evaluation uses pseudonymous speaker IDs (SPK-01–SPK-05), documented consent (`DATA_CONSENT_LOG.md`), restricted benchmark access, and PII-minimized metadata. Because voice recordings may themselves be identifying, raw audio requires controlled access and retention. Public reports do not expose private speaker names.

Scientific Disclosures & Known Limitations: Five explicit limitations are disclosed:

1. **Confidence Calibration:** The 30.9% ECE indicates weak confidence calibration under the heuristic classifier. It should not be interpreted as deployment-grade calibration.
2. **Speaker Equity Confound:** The performance difference between SPK-01 (90.9%) and SPK-02 (77.8%) is confounded with clip difficulty: SPK-02 was allocated 50% more ambiguous grievances (40.0% vs. 26.7%). It cannot be interpreted as an intrinsic voice disparity.
3. **Single-Labeler Baseline:** Tier A ground-truth references were established by a single expert labeler due to challenge timeline constraints.
4. **AfriSwitch Denominators & Provenance:** Upstream identifiers were added in v19 to eliminate transcript misalignment. Some Whisper subgroup WER values use reduced denominators after cross-script failures (51/60 scored overall).
5. **MSV Sample Composition:** MSV replicated the observed polarity pattern across three additional speakers. Its one-female/two-male composition is insufficient for gender fairness conclusions, and regional accent metadata remains unverified.

9. Reproducibility & Open Science Verification Suite

The complete benchmark suite is reproducible via the following commands executed from the repository root, all operating entirely offline from cached transcripts and test manifests without calling paid APIs:

- **Unit & Safety Tests:** `PYTHONPATH=backend python3 -m pytest backend/tests/` (validates 55 tests: gate logic, schema integrity, and raw transcript immutability).
- **Corpus Ingestion & v19 Baseline:** Verify frozen hash `sha256sum bench/results/v19_results.json` matches `c3cb688...` (full hash: `c3cb68824f70a231dd05f776de423042b02e750175c917e4367d82a108c127b8`); recomputed via `PYTHONPATH=backend python3 -m bench.metrics.run_full_benchmark`.
- **Multi-Speaker Validation (MSV):** `python3 bench/metrics/run_tier_a_multispeaker_validation.py` (clause-aware polarity adjudication across SPK-03, SPK-04, and SPK-05 from cached transcripts).
- **Polarity Stress Analysis:** `python3 bench/metrics/polarity_stress_analysis.py` (generates clip audit to `bench/results/tier_a_multispeaker_validation_audit.csv`).